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Leveraging the Chain on Goals Model in Football: Applications for Attack and Defensive Play

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Abstract: Introduction: Football analysis has experienced significant growth in recent years as an applied research field. This study aims to contribute to this area by applying the chain on goals model to analyze both the attacking and defensive phases of football matches. Additionally, it introduces four practical concepts to better understand player and team performance in Spain's professional football leagues. Method: Data for the 2023/24 season were collected from Football Reference, covering both men's (LaLiga) and women's (LigaF) leagues. Variables analyzed included team performance, attack and defensive performance, goals saved above average (GSAA), goals and possession value (PV), expected goals (xG), and xG on target (xGOT) for attack and defensive phases. Four practical concepts analyzed were off-ball movement (PV-xG), player's offensive quality (xG-xGOT), team's positioning (PVA-xGA), and player's defensive quality (xGA-xGOTA). Descriptive and comparative statistical analyses were performed to compare all variables between the two leagues using an Independent Student's test. Additionally, correlation coefficients were calculated to examine the relationships between the four concepts. Results: Significant differences were observed between leagues in defensive performance ($p = 0.03$) and GSAA ($p < 0.001$). Practical concepts revealed disparities in off-ball movement and team's positioning ($p < 0.001$ in both). No correlations were found between off-ball movement and player's offensive quality or between team's positioning and player's defensive quality. Conclusions: The Spanish women's league exhibited defensive weaknesses, conceding more goals and showing lower goalkeeper performance. PV was the most influential variable in the women's league, while xG was critical in the men's league.

Keywords: data analysis; football; expected goals on target; performance indicators

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1. Introduction

Football data analysis refers to the process of collecting, processing, and interpreting statistical information related to football matches, players, and teams to gain insights that can enhance performance, decision-making, and strategy. With advancements in technology and data science, football data analysis has become a crucial tool for clubs, coaches, and scouts to improve everything from game tactics to player development [1,2].

In the early days of football, the use of data was extremely limited, often revolving around simple metrics such as goals scored, assists, yellow and red cards, shots on target, or possession percentage [3]. Coaches relied on their experience and intuition, while perfor-

mance assessments were mostly subjective. Tactics were developed based on observation rather than quantifiable metrics [3].

In the 1990s, football saw a growing interest in statistical tracking. The introduction of companies like Opta in 1996 marked a turning point [4]. These companies began to record and analyze more detailed aspects of the game, such as passing accuracy, tackles and interceptions, distance covered, or number of crosses and corners [4,5]. These data allowed clubs to start making more informed decisions about player performances and match strategies.

The early 2010s marked the beginning of the Big Data revolution in football. Clubs began to accumulate vast amounts of data, not just from matches but also from training sessions. Simultaneously, the introduction of advanced statistical models revolutionized how performance was measured, as they highlighted the role of the shot, specifically the characteristics before and after the shot [6–10]. These included expected goals (xG) [11,12] or expected goals on target (xGOT) [13]. xG is a statistical metric used in football to measure the quality of scoring chances and the likelihood of a given shot resulting in a goal. The value of this metric assigned to a shot represents its "goal probability" and is determined by various factors such as the distance of the shot, the shooting angle, the type of shot (e.g., header or strike), the positioning of defenders, and the play sequence leading up to the attempt [14]. xG quantifies the likelihood of a shot resulting in a goal on a scale from 0 to 1, where a higher xG value indicates a greater probability of scoring [9,10]. In contrast, the expected goals on target (xGOT) metric offers a more detailed analysis by incorporating not only the quality of the scoring opportunity but also the execution of the shot itself. xGOT assesses the probability of a goal after the shot has been taken and is on target, considering factors such as shot placement, power, and the goalkeeper's positioning and reaction time. This refinement providers xGOT a deeper layer of analysis, focusing on the interplay between shooting quality and goalkeeping effectiveness [13].

Following this approach, in 2024, Ruiz-de-Alarcón-Quintero and De-la-Cruz-Torres [15] introduced an analysis model that emphasizes the process over the outcomes, aiming to offer more comprehensive insight into players' or teams' performance during a match or throughout season. This model, referred to as the "chain on goals model", differs from other models by incorporating multiple metrics instead of relying on a single one [6,11,12]. The model was structured around four primary metrics: goals scored, possession value (PV), xG, and xGOT. Each metric functions as a link in a chain, with the model premised on the idea that these interconnected data points collectively form a comprehensive representation of performance. Furthermore, the integrated analysis of these metrics offers opportunities to derive novel and insightful information relevant to football analytics. This chain on goals model was designed to provide a more precise evaluation of player and team performance in football matches. Using data from the Spanish men's and women's football leagues, the study [15] revealed that xG and xGOT were the metrics most strongly correlated with match outcomes, as they closely aligned with goals scored. Such insights enable coaches to identify the strengths and weaknesses of their teams, specifically before and after the shot.

To date, the scientific literature on xG and xGOT metrics is limited [9,11,16–18], and this remains the only article on the chain on goals model in football. More studies applying this model to aspects of football are needed to assess its broader potential. Therefore, the aim of this article was to provide a valuable contribution by applying the chain on goals model to both the attacking and defensive phases of a football team. Additionally, we introduced four practical concepts to enhance the understanding of player and team performance within the context of Spanish football leagues. Through this practical interpretation, the model could offer coaches a significant competitive advantage by enabling them to identify, with precision, the strengths and weaknesses of their players and teams. These insights

can be pinpointed across various situations: offensive or defensive phases, before or after shot on target, individual player quality, or overarching team strategies. By leveraging this model, coaches can drive meaningful improvements in performance.

2. Materials and Methods

2.1. Sample

The sample involves analyzing data from two Spanish professional football leagues, LigaF (women's first division) and LaLiga (men's first division), for the 2023–2024 season. The analysis covers 30 matchdays from LigaF and 38 matchdays from LaLiga, using data sourced from Football Reference (www.fbref.com (accessed on 20 October 2024)), which relies on datasets from Opta (Stats-Perform) (www.fotmob.com (accessed on 20 October 2024)). As far as the authors are aware, these datasets are predominantly inaccessible to the broader public [18,19]. Organizations like Opta and Stats Bomb independently gather these data and distribute their findings exclusively through proprietary platforms. The limited transparency, combined with the lack of insight into the algorithms employed to compute these metrics—a phenomenon often termed "black-boxing"—hampers the ability to fully understand and interpret the processes underlying the generation of specific metric values. However, they have been validated and are widely used in the football analytics community [20,21].

2.2. Design and Ethics Committee

A descriptive and comparative study was conducted. This involved performing a comparative analysis of the end-of-season aggregated match statistics collected over a competitive season from professional Spanish football teams in both the men's and women's leagues. The study received approval from the local ethics committee of University of Seville (protocol code 2024–1326 and date of approval 26 June 2024).

2.3. Methodology

The study variables were grouped into three dimensions: overall variables (team performance, attack performance, defensive performance, and goals saved above average), metrics from the chain on goals model (goals, PV, xG, and xGOT for the attack and defensive phase), and four practical concepts (off-ball movement and player's offensive quality for attack phase, and team's positioning and player's defensive quality for defensive phase). These concepts were developed by the authors, drawing on their extensive experience of over 25 years as data analysts in football. The first two metrics of the phases correspond to the team's strategies, whereas the latter two metrics provide insights into the individual performance of the players. The integrated analysis of these metrics provides opportunities to generate novel and insightful variables, offering greater complexity and relevance to football analytics. The authors proposed that these concepts represent a practical interpretation of the studied metrics, facilitating a deeper understanding for coaches in evaluating both player and team performance.

Table 1 presents the definitions of the variables for each dimension. According to experts in football analytics, all results for each variable from the first and second dimensions were normalized per match (90 min) to enhance understanding and interpretation. This standardization is essential in data analysis studies in football to enable comparison of data across different national leagues, as not all leagues have the same number of teams and, consequently, matches. This is also relevant for international competitions, where teams do not all play the same number of matches due to the varying structures of qualification phases.

Table 1. Definitions of the variables for each dimension.

Dimensions	Variables	Definitions
Overall	Team performance	Average points per match
	Attack performance	Average of goals scored per match
	Defensive performance	Average of goals conceded per match
	Goals saved above average (GSAA) (xGOTA-defensive performance)	Goals expected to be conceded by goalkeeper (GK) minus actual goals conceded
Chain on goals model	Goals	The total number of goals scored by the team per match during the season
	Possession value (PV)	The probability of a team scoring from their possession during the attack phase
	Expected goals (xG)	The quality of a shot based on historical data, providing an estimate of how likely a shot is to result in a goal during the attack phase.
	Expected goals on target (xGOT)	The post-shot quality of on-target efforts at goal during the attack phase
	Possession value against (PVA)	The probability that a team will allow itself to be scored on during its defensive phase
	Expected goals against (xGA)	The probability of a conceded shot to become a goal during your defensive phase.
	Expected goals on target against (xGOTA):	The post-shot quality of the shot attempts conceded during its defensive phase
Practical concepts	Off-ball movement (PV-xG)	The ability of players not carrying the ball from the attacking team to offer a passing solution to the teammate.
	Player's offensive quality (xG-xGOT)	The ability of a shooter to shot on target become a goal avoiding goalkeeper opposition
	Team's positioning (PVA-xGA)	The actions performed by the players of the defending team on space against their opponents without showing any kind of marking.
	Player's defensive quality (xGA-xGOTA)	The ability of a defender player to prevent a goal once that the attacker shoot

2.4. Statistical Analysis

The data were statistically analyzed using SPSS (version 18; SPSS Inc., Chicago, IL, USA). The Shapiro–Wilk test was used to perform normality analysis, and the result indicated a normal distribution. The statistical analysis performed a descriptive study of all variables expressed as the average and the standard deviations, as well as the percentiles (30th and 70th for LaLiga and 20th and 80th for LigaF). The difference in the calculation of the percentiles between the men's and women's football leagues is due to the fact that the women league has fewer teams and fewer competitions during the season.

Pearson correlation coefficient (r) compared the interactions between variables and R2 analyzed the correlation between off-the-ball movement and player's offensive quality and team's positioning and player's defensive quality. This investigation applied the following criteria to determine the magnitude of the correlation (r): <0.1 trivial, 0.1 to 0.3 small, 0.3 to 0.5 moderate, 0.5 to 0.7 large, 0.7 to 0.9 very large, and 0.9 to 1.0 almost perfect.

Finally, a comparative study was conducted to compare all variables between both leagues, using Independent Student's *t*-test. Statistical significance was set at $p < 0.05$. Effect sizes (Cohen's d) were also calculated to assess whether the significance findings also held practical significance (small < 0.2; medium 0.5; and large > 0.8).

3. Results

Considering the definitions of the variables presented in Tables 1 and 2 illustrates the distribution of values that were calculated for all variables mentioned along the entire season, including mean, standard deviation, and percentile distribution, for both leagues. In the overall dimension only, defensive performance (measured as the average number of goals conceded per match) and GSAA showed statistically significant differences between the two leagues ($p = 0.03$ and d= −0.57; $p < 0.001$ and d = −1.26, respectively).

Table 2. The values for the variables of the overall dimension and chain in goals model metrics in each league. *p* value for between-groups statistical significance (Independent Student's *t*-test) are reported.

Variable	LigaF	LaLiga	*p* Value	LigaF		LaLiga	
	Mean ± SD	Mean ± SD		Weakness Index (20th)	Strength Index (80th)	Weakness Index (30th)	Strength Index (70th)
Performance	1.41 ± 0.68	1.36 ± 0.54	0.63	0.94	2.02	1.05	1.56
Offensive performance	1.58 ± 0.95	1.32 ± 0.48	0.58	1.05	2.02	1.00	1.53
Defensive performance	−1.57 ± 0.55	−1.32 ± 0.33	*0.03*	−1.93	−0.97	−1.45	−1.17
GSAA	−0.19 ± 0.19	0.00 ± 0.11	*<0.001*	−0.36	0.04	−0.08	0.06
PV	1.25 ± 0.67	1.34 ± 0.28	0.39	0.84	1.52	1.14	1.50
xG	1.39 ± 0.74	1.30 ± 0.31	0.83	0.95	1.69	1.12	1.37
xGOT	1.35 ± 0.67	1.32 ± 0.34	0.85	0.95	1.74	1.12	1.50
PVA	−1.25 ± 0.33	−1.34 ± 0.18	0.50	−1.55	−1.03	−1.42	−1.24
xGA	−1.39 ± 0.40	−1.30 ± 0.21	0.27	−1.64	−0.97	−1.434	−1.14
xGOTA	−1.38 ± 0.42	−1.32 ± 0.25	0.30	−1.60	−0.93	−1.41	−1.14

Abbreviations: GSAA, goals saved above average; PV, possession value; xG, expected goals; xGOT, expected goals on target; SD, standard deviation.

Table 3 presents the reference values for the four practical concepts in each league. Significant differences were observed only in off-ball movement ($p < 0.001$ and d = 1.31) and team's positioning ($p < 0.001$ and d = −1.34) when comparing the two leagues.

Table 3. The values for the four practical concepts in each league. *p* value for between-groups statistical significance (Independent Student's *t*-test) are reported.

Variable	LigaF	LaLiga	*p* Value	LigaF		LaLiga	
	Mean ± SD	Mean ± SD		Weakness Index (20th)	Strength Index (80th)	Weakness Index (30th)	Strength Index (70th)
Off-ball movement	0.14 ± 0.12	−0.04 ± 0.15	*<0.001*	0.05	0.24	−0.09	0.01
Player's offensive quality	−0.04 ± 0.17	0.02 ± 0.11	0.13	−0.17	0.10	−0.08	0.10
Team's positioning	−0.14 ± 0.14	0.04 ± 0.13	*<0.001*	−0.25	0.00	−0.06	0.12
Player's defensive quality	0.01 ± 0.08	−0.02 ± 0.09	0.93	−0.07	0.05	−0.05	0.02

For LigaF, the correlation coefficients between off-the-ball movement and player's offensive quality were r = −0.1207 and R2 = 0.0146 and between team's positioning and player's defensive quality were r = −0.1074 and R2 = 0.0115. For LaLiga, the correlation coefficients between off-the-ball movement and player's offensive quality were r = −0.1324 and R2 = 0.0175 and between team positioning and player's defensive quality were r = −0.0280 and R2 = 0.008. In all cases, the level of correlation was trivial. This implied that the practical concepts of each phase of the game could be considered independent of one another, ensuring that the weakness or strength of one phase did not influence the others.

Figures 1 and 2 establish the relationship of off-ball movement and player's offensive quality and team's positioning and player's defensive quality for all teams in the women's and men's leagues, respectively. From these figures, the strengths and weaknesses of each team can be identified, whether in the defensive or attack phase, at any given point in time or at the conclusion of the season, as demonstrated in this study. In the defensive evaluation figures, the x-coordinate represents the team's positioning, while the y-coordinate represents the player's defensive quality. Therefore, a team's position further to the right on the x-coordinate indicates better team's positioning, while a position further to the left reflects a lower positioning value. On the y-coordinate, an upward shift corresponds to higher player defensive quality, whereas a downward shift indicates lower player defensive quality. In the attack evaluation figures, the x-coordinate represents the off-ball movement, while the y-coordinate represents the player's offensive quality. Thus, greater off-ball movement is indicated by a position further to the right on the *x*-axis, whereas a position further to the left reflects lower off-ball movement. On the *y*-axis, an upward shift corresponds to higher player offensive quality, whereas a downward shift indicates lower player offensive quality.

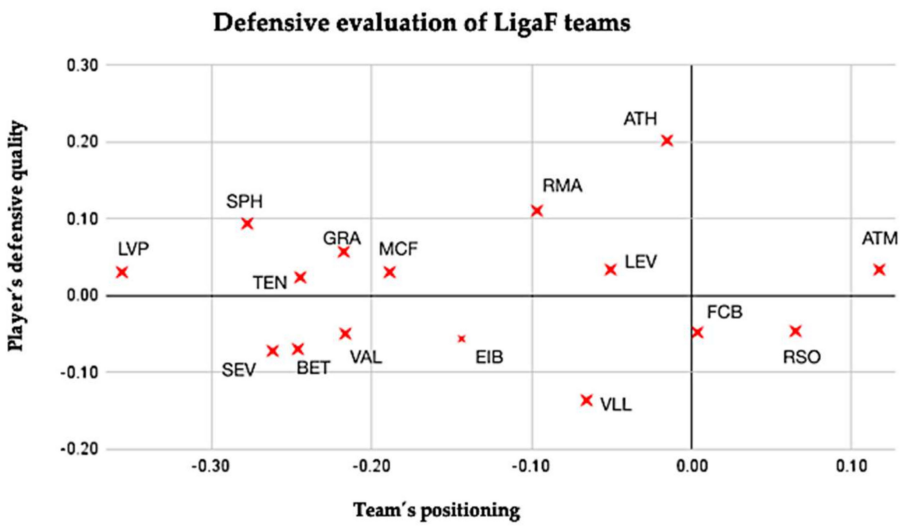


Figure 1. *Cont.*

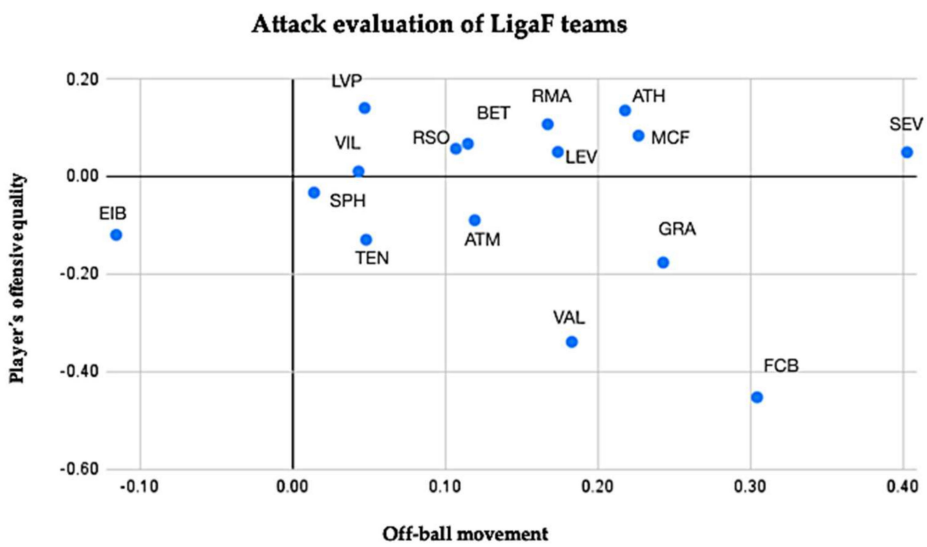
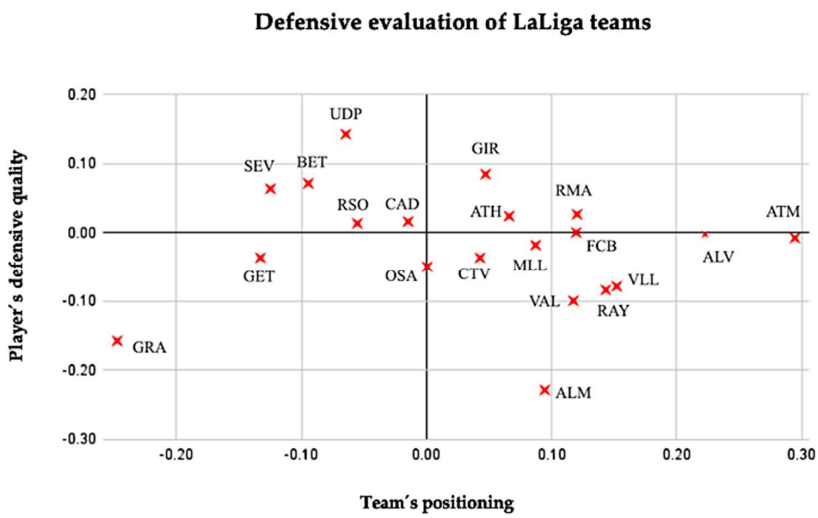


Figure 1. The relationship between the team's positioning with player's defensive quality (**above**) and off-ball movement with player's offensive quality (**below**) for each team in the women's league.



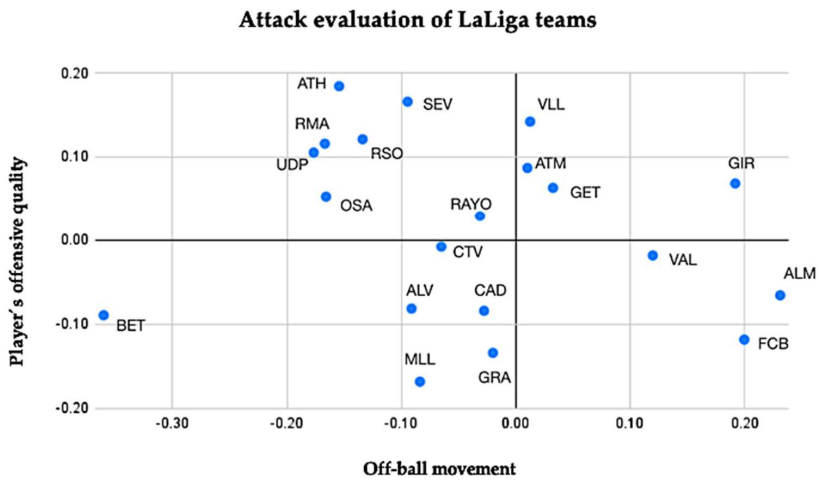


Figure 2. The relationship between team's positioning with player's defensive quality (**above**) and off-ball movement with player's offensive quality (**below**) for each team in the men's league.

4. Discussion

The data from this study offer valuable experimental evidence that can effectively inform the performance of football players and teams. With regard to the variables presented in Table 1, the primary finding was the establishment of reference values for the variables that constitute the chain on goals model in football (Table 2), applicable to the attack and defensive phases of teams in both the Spanish women's and men's football leagues. Notably, there is similarity between the two leagues. However, in the overall dimension, the defensive performance (average number of goals conceded per match) and GSAA were statistically different when comparing both leagues ($p = 0.03$ and $p < 0.001$, respectively) (Table 2). These significant differences had a medium and large effect size, respectively. Regarding defensive performance, the authors concluded that more goals were conceded in the women's league, indicating a weaker defensive system in this league. Concerning GSAA, this metric analyzes the balance between GK performance and shooter performance. A positive GSAA outcome indicates that the GK's performance is better than the shooter's, whereas a negative outcome suggests the opposite. If the GSAA value is "0", it signifies a neutral balance between GK and shooter [22]. Our data revealed that the men's league was balanced, as there was an equilibrium between the performance of the shooter and the GK. In contrast, the women's league exhibited an imbalance, suggesting a poorer performance by the GK.

Concerning the second objective, the authors sought to present four practical concepts based on the metrics of the chain on goals model, aimed at assisting coaches and technical staff in better understanding a team's tactical performance and that of its players (Table 3). Figures 1 and 2 illustrated the analysis and interrelationships of these four concepts across all professional teams in the Spanish league, both women's and men's, during 2023–2024 season. One of the primary responsibilities of analysts is to thoroughly examine the data to review performance metrics, enabling coaching staff to pinpoint areas for improvement, such as missed opportunities, defensive weaknesses, or necessary tactical adjustments [23]. Therefore, by analyzing these concepts, we can discern whether football errors stem from the team's strategy or from individual player quality. Specifically, the concepts of "off-ball movement" and "team's positioning" provide insights into the team's strategy, while "player's offensive quality" and "player's defensive quality" relate to the quality of the individual players. Notably, the concepts defining the defensive and attack phases showed a low correlation. The authors suggest that this discrepancy arises because "off-ball movement" and "team positioning" reflect the overall strategy of the team, whereas "player's offensive quality" and "player's defensive quality" are indicative of individual player quality. As a result, the two sets of concepts operate independently from each other. Curiously, the two metrics reflecting information about team strategy differed significantly between the two leagues ($p < 0.001$ in both cases), with a large effect size in both cases. Regarding off-movement, the data indicated that PV was more decisive in the women's league, while xG was more critical in the men's league. This finding aligns with a previous study [15], which found that the predictive value of PV in relation to match outcomes in the Spanish women's league showed a stronger correlation than in the men's league. Concerning team's positioning, the data revealed that, from a collective perspective, the men's league demonstrated better defensive positioning than the women's league. The authors hypothesize that, while the football pitch dimensions are the same for both leagues, the physical dimensions of men and women players differ. Consequently, the ability to cover the playing field during the defensive phase may be greater in the men's league.

Based on these findings, the differences between leagues identified in the study may be attributed to the limited resources that women's leagues continue to face, including

insufficient funding, variations in style of play, physical preparation, and other factors, all of which can influence the level of competition.

The analysis of football data has progressed from basic statistics to a powerful tool that impacts every aspect of the game, including team and player performance evaluation as well as in-match decision-making [1]. This is why, while previous studies have compared successful (winning) and unsuccessful (losing) football teams [24–26], this analysis offers a novel approach by incorporating more advanced metrics. This allows for a more in-depth evaluation of the factors associated with sporting success. This study provides valuable reference values from elite football clubs that can aid in shaping tactical strategies (Tables 2 and 3). Similar to a previous study [15], this article establishes reference values for model metrics based on data from top-tier clubs, such as those in the Spanish men's and women's leagues. The authors also provide a percentile distribution to indicate reference values considered as strength indicators (p70 for the men's league and p80 for the women's league), capturing scores for teams competing at the European level by the final matchday. Conversely, values identified as indicators of weakness (p30 for the men's league and p20 for the women's league) represent scores for teams in relegation positions to a lower-tier league by the last matchday (Tables 2 and 3).

4.1. Limitation Section

There are several limitations to this study. (1) The data used in this research come from one of Europe's major leagues, the Spanish soccer league. Therefore, these data should be considered indicative and cannot be generalized. Applying this metric to other European leagues or on a global scale could prove to be insightful; (2) The results of this study were derived from a single data provider; however, the data have been validated and are widely utilized within the football analytics community [20,21]; (3) The scientific literature regarding this model and its metrics is quite limited. Further research is required to apply these new metrics to technical and tactical strategies, which would be highly beneficial for coaches and technical staff; (4) Today, the information obtained from the application of the chain on goals model is based on detecting strengths and weaknesses in football teams. Studies that relate this information with football solutions for improving the performance of teams and players are necessary; (5) The study was performed using a static method, as opposed to a dynamic approach. Nevertheless, this static method may prove to be pedagogically valuable for coaches in decision-making during matches [23].

4.2. Practical Application

The data reveal patterns of play that help coaches refine tactics, such as exploiting weaknesses in the opponent's formation, capitalizing on space created by opposing defenders, and enhancing players' skills to boost their performance. In fact, the Spanish men's league should prioritize the development of its forwards' talent, just as the women's league should focus on enhancing the skills of its GKs.

A detailed analysis of the teams allows the authors to raise the following reflections: In LigaF, Sevilla FC (SEV) recorded the lowest scores for both team's positioning and player's defensive quality, indicating poor defensive organization and limited ability of defenders to prevent goal-scoring opportunities. Coaches should focus on improving both the overall defensive structure and the individual skills of defenders to better prevent shots on goal. Conversely, Barcelona FC (FBC) achieved a high score for off-ball movement but a low score for player's offensive quality, suggesting that the team creates numerous goal-scoring opportunities but struggles to convert them into goals. Coaches should prioritize enhancing the shooting skills of shooters.

In LaLiga, Atlético de Madrid (ATM) exhibited a high team's positioning value and a standard player's defensive quality value, which suggests a strong defensive unit and a team that concedes few goals. On the other hand, FC Barcelona (FCB) recorded the best off-ball movement score but a low player's offensive quality score, indicating that, despite generating numerous scoring opportunities, the team struggles to capitalize on them. Coaches should focus on strengthening the shooting abilities of shooters to improve goal conversion.

5. Conclusions

The Spanish women's league exhibited defensive weaknesses, conceding more goals and showing lower GK's performance. PV was the most influential variable in the women's league, while xG was critical in the men's league. Authors suggest that applying the chain on goals model, through the analysis of its metrics, may enhance the understanding of different phases of the game (both attack and defensive) in football teams. Furthermore, utilizing the four practical concepts enables coaches to conduct a comprehensive analysis of their tactical strategies and assess the quality of their players.

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Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki, and approved by the Ethics Committee of University of Seville (protocol code 2024-1326 and date of approval 26 June 2024).

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The original data presented in the study are openly available from the following website: www.fbref.com (accessed on 20 October 2024).

Conflicts of Interest: The authors declare no conflicts of interest.

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